

SOFTWARE ENGINEERING & PROJECT MANAGEMENT

MediTrack EMR

A unified, auditable Electronic Medical Record system for Aarogya Healthcare

Case Study 1 | Final Project Submission

B.Tech CSE 2025-29 | Semester III | School of Future Tech

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Companion documents	BRD, SRS, UML package, Estimation Report, Project Plan, Test Plan, Risk Register (submitted separately)

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1. Executive summary

Aarogya Healthcare runs 6 hospitals and 21 clinics across three states. Every site keeps patient history in a different way: a 14-year-old FoxPro package, unencrypted Excel registers, or paper files carried between sites. In 2024 this fragmentation produced **310 prescribing incidents traced to missing allergy history**.

MediTrack EMR replaces all of that with one system that holds registration, OPD/IPD notes, e-prescriptions, lab orders, pharmacy dispensing and discharge summaries, with a legally defensible audit trail. The system is designed around three non-negotiable demands from management:

- **Safety:** a prescription that conflicts with a recorded allergy must be impossible to sign.
- **Usability:** a nurse must be productive after 20 minutes of training.
- **Schedule:** delivery within a hard ceiling of 18 months.

What is original in this submission

- 1. A **hard-stop allergy interlock** driven by a 14-rule decision table.
- 2. **Billing fully decoupled** from prescribing using a Port-and-Adapter design and an AMQP message broker.
- 3. A **tamper-evident audit ledger** built from chained SHA-256 hashes.
- 4. A **12-stage migration pipeline** that quarantines the roughly 7% corrupted legacy rows instead of loading them.

2. The problem we were asked to solve

2.1 Organisation and context

Organisation	Aarogya Healthcare: 6 hospitals, 21 clinics, three states (27 sites)
Domain	Regulated clinical workflow: Electronic Medical Records
Current systems	Two sites on a 14-year-old FoxPro package, three on Excel registers, one with no digital record, one clinic using paper files that travel to the main hospital
Trigger event	310 prescribing incidents in 2024 traced to missing allergy history

2.2 Management mandate

Management wants **one EMR** that holds registration, OPD/IPD notes, e-prescriptions, lab orders, pharmacy dispensing and discharge summaries, with strict audit trails because clinical records are legally admissible and cannot be silently edited.

Stakeholder	Constraint they imposed
Chief Medical Officer	The system must never lose a record, and a nurse must be able to use it after 20 minutes of training.
Chief Financial Officer	The project cannot exceed 18 months.
Legal / Compliance	Records must be audit-ready and tamper-evident; nothing may be silently edited.

2.3 Given project data (from the case study)

Parameter	Value
Estimated size	46 KLOC
Team	Medium-experienced; regulated clinical workflow
Legacy migration	2.4 million patient rows; roughly 7% have malformed dates or duplicate IDs
Defect removal efficiency target	At least 92% before go-live
Reliability target	MTBF of 1,400 hours against an MTTR of 2 hours
Schedule ceiling	18 months

2.4 Why this is hard

- **Safety-critical:** a software defect can harm a patient, so correctness matters more than speed of delivery.
- **Legal:** records are evidence. Tampering must be detectable, not merely discouraged.
- **Messy legacy data:** 2.4 million rows from three incompatible sources, with a known corrupt fraction.
- **Human factors:** nurses are busy and under-trained on software; the UI cannot rely on manuals.

3. Process model: why the V-Model

This project follows the **Regulated V-Model (Verification and Validation lifecycle)**. Each development phase on the left of the V is paired with a test phase on the right, and the test cases are written at the same time as the requirement they verify, before any code exists.

Model	Why it fits poorly	What the V-Model does instead
Pure Waterfall	Testing arrives at the very end, so a safety defect found late forces expensive rework against a hard 18-month ceiling.	Test design runs in parallel with every phase, so defects in requirements are found while they are cheap.
Pure Scrum	Favours working software over comprehensive documentation; regulated medical software needs documented requirements, traceability and formal sign-off.	Keeps full documentation and a traceability matrix, while still allowing iterative UI feedback.
V-Model	Fits fixed, well-understood, safety-critical scope.	Every requirement has a test; every test traces back to a requirement.

Worked example: the safety interlock (FR-07)

FR-07 states that a prescription conflicting with a recorded allergy cannot be signed. Its Boundary Value Analysis tests and its Decision Table tests were designed while the requirement was being written, so the requirement was already testable before coding began. This is the V-Model idea in one sentence: requirement and test are born together.

The approach is consistent with the principles of IEC 62304 (medical device software lifecycle) and with HIPAA-style privacy and audit expectations.

4. The solution at a glance

The table below lists the main capabilities, and, more importantly, **why** each one exists. Each feature traces back to a stated problem rather than being added for its own sake.

Feature	What it does	Why it exists
Unified patient record	One record per patient across all 27 sites, replacing FoxPro, Excel and paper.	Fragmentation is the root cause of the 310 incidents.
Structured allergy history	Allergies captured at registration/triage with substance, drug class and severity.	The safety engine can only block what it can see.
OPD/IPD clinical notes	Encounter-based notes by doctors and nurses.	Mandated by management as part of the single record.
E-prescribing (CPOE)	Doctor writes prescriptions electronically with live safety checks.	Removes handwriting errors and enables automated safety checks.
Allergy hard-stop interlock	Blocks signing when a drug conflicts with an allergy, including cross-class (e.g. Penicillin and Amoxicillin).	Directly targets the 310 incidents.
Lab orders and pharmacy dispensing	Orders and dispensing recorded against the same patient and encounter.	Closes the loop from prescription to dispensing.
Discharge summary	Generated from the encounter record.	Required by management; reduces re-typing errors.
Role-based access control	Doctors, nurses, pharmacists and admins see only what their role needs.	Privacy and least-privilege for regulated data.
Cryptographic audit trail	Every clinical event written to an append-only, hash-chained ledger.	Legal admissibility; no silent edits.
Nurse-first interface	Large targets, clear colour coding, minimal screens per task.	The 20-minute training mandate.

5. Extra features beyond the brief

The case study asks for a working design that meets the stated constraints. The four items below go further. For each one: what it is, why it was needed, and what it buys the hospital.

5.1 Hard-stop safety interlock (14-rule decision table)

What	A rule engine that checks every drug in a draft prescription against the patient's allergy record. If a conflict is found, the Sign and Submit control is disabled at the interface and the server rejects the request as well. The 14 rules cover direct matches and cross-class reactions (for example a Penicillin allergy blocking Amoxicillin because both are beta-lactams).
Why	A warning that a tired clinician can click through is not a safeguard. The 310 incidents happened because information was missing; the next risk is information being present but ignored.
Benefit	The class of error behind the 2024 incidents becomes structurally impossible to commit, not just discouraged.

Verification	The decision table doubles as the test oracle: each rule is one test case in the Test Plan.
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Alert tiers in the prototype. Not every conflict is treated the same way. The dashboard's interception feed shows three outcomes: a **HARD STOP** (for example Amoxicillin for a Penicillin-allergic patient, or an NSAID for an asthmatic patient), a **Cross-Reactivity Alert** that needs clinician attention (for example Cefazolin), and a controlled **Override** for a clinical-parameter case (for example Gentamicin with renal clearance). Anaphylaxis-level allergies are always hard stops.

Illustrative excerpt of how a decision table reads (the full 14-rule table is in the Test Plan):

Rule	Condition: allergy recorded for	Condition: drug class	Severity	Action
R1	Penicillin	Penicillin (same class)	Any	HARD STOP
R2	Penicillin	Other beta-lactam (cross-class)	Severe	HARD STOP
R3	No allergy on record	Any	n/a	PASS
R4	Allergy field not yet captured	Any	n/a	BLOCK until allergy status recorded
R5	Unrelated class	Different class	Any	PASS

5.2 Decoupled billing (Port-and-Adapter with an AMQP broker)

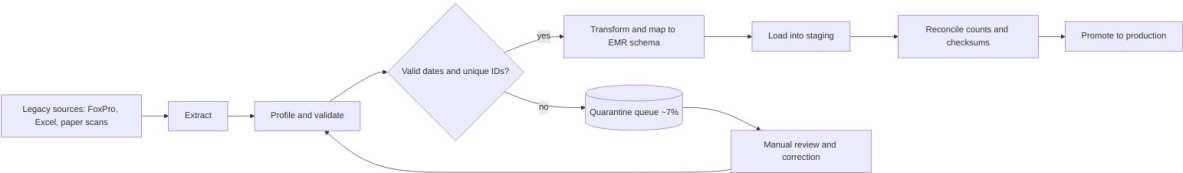
What	The Prescription Service never calls billing. After it commits the prescription it publishes a PrescriptionIssuedEvent to an AMQP message broker. A Billing Adapter subscribes to that event and creates the charge in the external billing system.
Why	Case-study objective: keep the prescription module loosely coupled from billing. Billing is a commercial back-office function; clinical care must not wait for it.
Benefit	If billing is slow or down, doctors keep prescribing. Billing catches up from the queue when it recovers. Replacing the billing vendor means writing one new adapter, with no change to clinical code.

5.3 Cryptographic audit trail (chained SHA-256)

What	Each audit entry stores the hash of the previous entry together with its own content, and the ledger is append-only. Changing any historical entry changes its hash and breaks every hash after it.
Why	Clinical records are legal evidence and must not be silently edited. A normal database log can be edited by a privileged user without anyone noticing.
Benefit	Tampering becomes detectable by a simple re-computation, which supports admissibility. The ledger is tamper-evident; it is not claimed to be tamper-proof.

5.4 Legacy migration quarantine (12-stage ETL pipeline)

What	A staged extract-validate-transform-load pipeline for 2.4 million legacy rows. Rows with malformed dates or duplicate IDs (about 7%, roughly 168,000 rows) are diverted to a quarantine queue for review instead of being loaded or silently discarded.
Why	Loading dirty data would poison the new system; deleting it would violate the "never lose a record" rule from the CMO.
Benefit	The production database contains only validated rows, and every quarantined row remains traceable and recoverable. A regression suite re-checks counts and checksums after each load.



Migration flow with quarantine (simplified view of the 12 stages)

6. User interface prototype

A high-fidelity single-page application prototype demonstrates the 20-minute onboarding mandate. Every screen follows the same layout: a dark navy sidebar for navigation (grouped into **Clinical Core**, **Governance & Operations** and **Engineering Deliverables**), a header showing the connected facility and live availability, and a white work area with one task per screen. Red is used for safety and critical states, green for safe or verified states, so colour itself tells the user what to do.

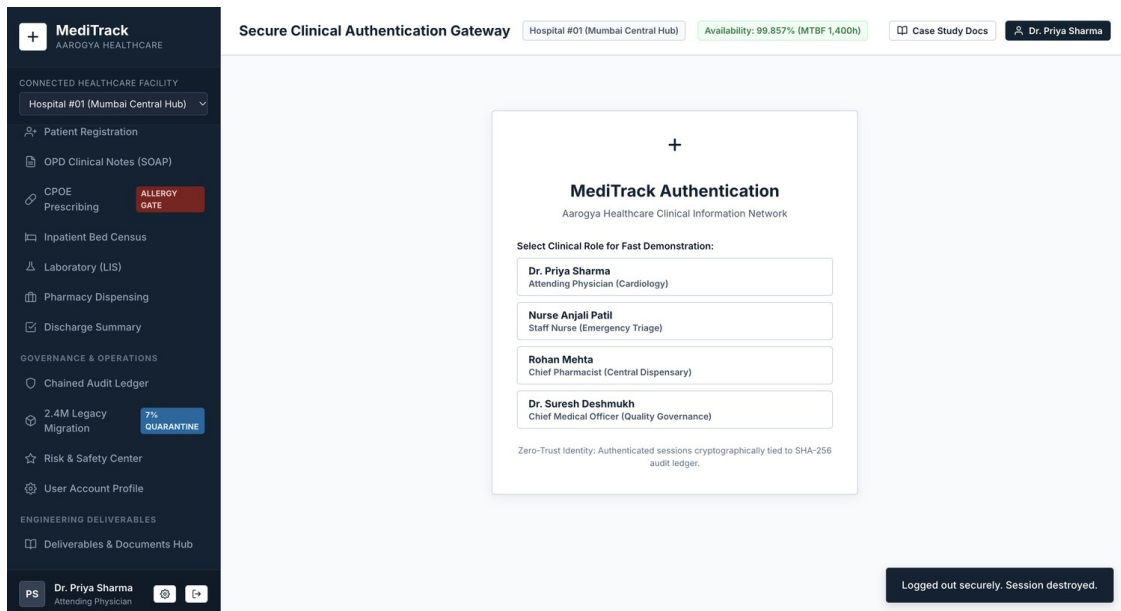
6.1 Design decisions and the reason for each

Decision	Reason
Same layout on every screen	A nurse who learns one screen has learned them all; this supports the 20-minute training target.
One task per screen, grouped sidebar	Reduces what must be remembered; clinical tasks sit apart from governance tasks.
"ALLERGY GATE" badge on the CPOE menu item	Reminds every prescriber that the safety interlock is always active.
Allergy shown on the patient banner and in red	The most dangerous missing information is always in view, which targets the root cause of the 310 incidents.
Red for critical, green for safe	Consistent colour meaning prevents misreading under time pressure.
Facility selector and availability chip in the header	Shows which of the 27 sites the user is working in, and the live reliability target (MTBF 1,400 h).

6.2 Screen-by-screen walkthrough

The screens below follow the order in which a patient moves through the hospital. Each one lists what it shows and which requirement or extra feature it demonstrates.

Secure sign-in with role selection

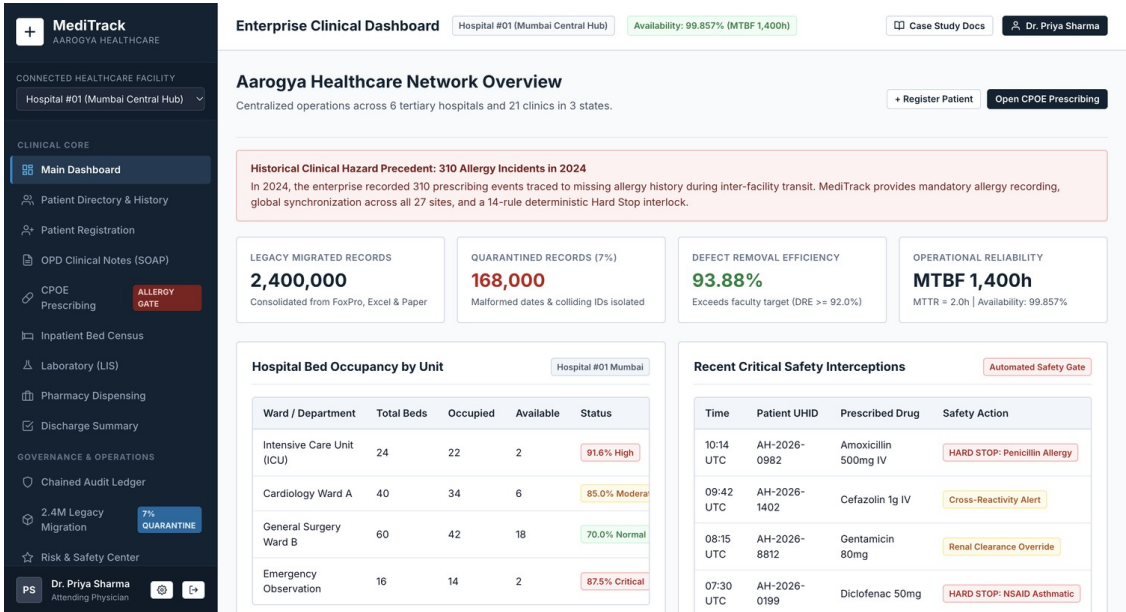


Secure sign-in with role selection

What it shows: The authentication gateway with four demonstration roles: attending physician, staff nurse, chief pharmacist and chief medical officer. Sessions are tied to the audit ledger.

Why it matters: Role-based access control and traceability: every later action is attributed to a verified person. The role buttons make the demo fast without weakening the design.

Enterprise clinical dashboard

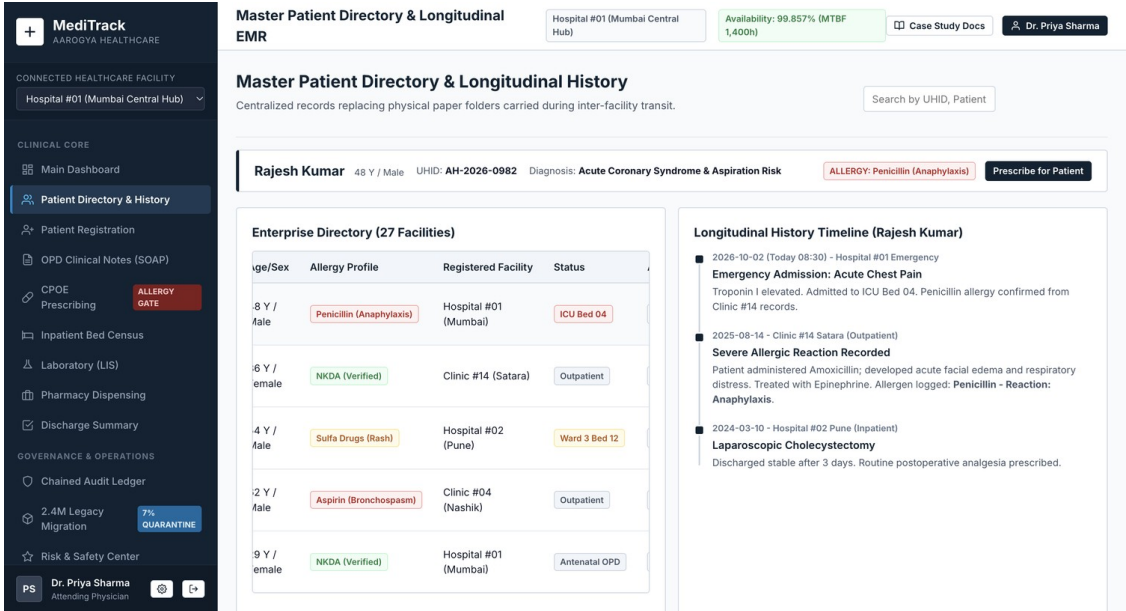


Enterprise clinical dashboard

What it shows: The network overview for the 6 hospitals and 21 clinics. A red banner restates the 2024 problem (310 allergy incidents). Four tiles show 2,400,000 migrated records, 168,000 quarantined (7%), Defect Removal Efficiency of 93.88% against the 92% target, and MTBF 1,400 h with 99.857% availability. Below are bed occupancy by unit and a live feed of recent safety interceptions (hard stops, cross-reactivity alerts, an override).

Why it matters: It puts the project's measurable targets and the safety record in one place, so management can see at a glance that the system meets its acceptance criteria.

Master patient directory and longitudinal history



Master patient directory and longitudinal history

What it shows: One searchable directory across all 27 facilities with allergy profile, registered facility and status for each patient. For the selected patient, Rajesh Kumar, a timeline shows a severe Penicillin reaction recorded at Clinic #14 in 2025, a surgery at Hospital #02 in 2024 and today's emergency admission at Hospital #01.

Why it matters: This is the direct answer to fragmentation. The allergy recorded at a small clinic in Satara is visible to the Mumbai emergency team, which is exactly the information that was missing in the 310 incidents.

Patient registration with mandatory allergy gate

MediTrack
AAROGYA HEALTHCARE

CONNECTED HEALTHCARE FACILITY
Hospital #01 (Mumbai Central Hub)

CLINICAL CORE

- Main Dashboard
- Patient Directory & History
- Patient Registration**
- OPD Clinical Notes (SOAP)
- CPOE Prescribing **ALLERGY GATE**
- Inpatient Bed Census
- Laboratory (LIS)
- Pharmacy Dispensing
- Discharge Summary

GOVERNANCE & OPERATIONS

- Chained Audit Ledger
- 2.4M Legacy Migration **7% QUARANTINE**
- Risk & Safety Center

PS Dr. Priya Sharma
Attending Physician

New Patient Registration (Mandatory Allergy Gate)

Hospital #01 (Mumbai Central Hub) Availability: 99.857% (MTBF 1,400h) Case Study Docs Dr. Priya Sharma

New Patient Registration & Master Indexing

Mandatory allergy verification gate enforced at registration in compliance with CMO safety protocol.

First Name * Last Name *

e.g. Anand e.g. Kulkarni

Date of Birth (ISO 8601) * Biological Sex * Mobile Number

dd/mm/yyyy Male +91 98200 12345

Mandatory Clinical Safety Check: Drug Allergy Declaration

Registration staff must ask patient directly: "Have you ever had an allergic reaction, swelling, rash, or breathing difficulty after taking any antibiotic or medicine?"

Allergy Status *

No Known Drug Allergies (NKDA) - Verified

Cancel Complete Registration & Assign UHID

Patient registration with mandatory allergy gate

What it shows: The registration form cannot be completed without an explicit allergy status (for example "No Known Drug Allergies - Verified"). The staff member is prompted to ask the patient directly about reactions to any medicine.

Why it matters: The safety engine can only block what it knows. Making allergy status mandatory at the first point of contact removes the "unknown allergy" gap at its source.

OPD clinical consultation (SOAP notes)

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AAROGYA HEALTHCARE

CONNECTED HEALTHCARE FACILITY
Hospital #01 (Mumbai Central Hub)

CLINICAL CORE

- Main Dashboard
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GOVERNANCE & OPERATIONS

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PS Dr. Priya Sharma
Attending Physician

Outpatient Clinical Consultation (SOAP Notes)

Hospital #01 (Mumbai Central Hub) Availability: 99.857% (MTBF 1,400h) Case Study Docs Dr. Priya Sharma

Outpatient Clinical Consultation (SOAP Protocol)

Signed electronic encounter documentation satisfying legal medical admissibility standards. Consulting: Rajesh Kumar (AH-2026-0982)

Subjective (Chief Complaint & Patient History) *

Patient presents with acute substernal chest discomfort radiating to left arm. Onset 2 hours ago. Dyspnea on minimal exertion. Reports known Penicillin allergy.

Objective (Vital Signs & Physical Examination)

BP: 144/92 mmHg | HR: 88 bpm | SpO2: 97% on room air | Temp: 36.8 C. Cardiovascular: S1 S2 heard, no murmurs. Lungs: Clear bilaterally, no crepitations.

Assessment (Clinical Impression & ICD-10 Diagnosis) *

I20.0 - Unstable Angina / Acute Coronary Syndrome (Rule Out NSTEMI)

Plan (Investigations & Treatment Directives)

1. Stat 12-lead ECG. 2. Serial Troponin I at 0 and 3 hours. 3. Transfer to Emergency Observation Unit. 4. CPOE prescription for anti-ischemic regimen (Avoid Beta-Lactam antibiotics).

Author: Dr. Priya Sharma (MH-MED-2018-0914) | Legally Admissible Signature

Save & Sign Clinical Note

OPD clinical consultation (SOAP notes)

What it shows: A structured note with Subjective, Objective, Assessment (ICD-10 coded) and Plan sections, signed by the doctor with their licence number. The plan itself says to avoid beta-lactam antibiotics for this patient.

Why it matters: Covers the OPD/IPD notes requirement and produces a legally admissible signed record. The structured format also keeps notes consistent across sites.

CPOE prescribing with allergy interlock

CPOE Prescribing & Allergy Interlock Hospital #01 (Mumbai Central Hub) Availability: 99.857% (MTBF 1,400h) Case Study Docs Dr. Priya Sharma

Computerized Physician Order Entry (CPOE) Prescribing
Integrated 14-rule safety interlock intercepting historical allergy failure modes. Patient Allergen: Penicillin (Anaphylaxis)

Rajesh Kumar UHID: AH-2026-0982 VERIFIED FATAL ALLERGY: PENICILLIN Test Prescribing Amoxicillin (Allergy Hazard)

Safety Gate Cleared: No Allergy or Drug Interaction Detected
Selected drug class (Macrolide: Azithromycin) has 0% Beta-Lactam ring cross-reactivity with documented Penicillin sensitivity. Order is safe to proceed.

Electronic Prescription Entry
Medication / Active Substance *: Azithromycin 500mg Oral Tablet (Macrolide - Safe for Penicillin Allergy)
Dose per Administration (mg) *: 500 Dosing Frequency *: Once Daily (OD)
Duration (Days) *: 5
BVA Verified: Therapeutic window valid (125 mg to 1000 mg).
Simulate Allergy Hazard Sign & Authorize E-Prescription

Active Authorized Medications (Rajesh Kumar)

Rx ID	Drug	Dose	Freq	Dur	Status	Pr
RX-2026-7841	Azithromycin 500mg Oral	500 mg	Once Daily	5 Days	Active / Signed	Dr. Priya Sharma
RX-2026-7102	Atorvastatin 40mg Oral	40 mg	Nightly	30 Days	Active / Signed	Dr. Priya Sharma
RX-2026-6990	Aspirin 75mg Oral	75 mg	Once Daily	30 Days	Active / Signed	Dr. Priya Sharma

CPOE prescribing with allergy interlock

What it shows: The e-prescription form for the penicillin-allergic patient. Azithromycin (a macrolide) is selected, so a green banner explains it has no beta-lactam cross-reactivity and the order may proceed. The dose field shows a Boundary Value Analysis check (valid window 125 to 1000 mg). The "Simulate Allergy Hazard" button demonstrates the red hard stop by trying Amoxicillin.

Why it matters: This is the central screen of the project: the interlock (extra feature 5.1), the BVA dose validation from the test plan, and the sequence diagram in section 7.4 all appear here.

Inpatient bed census and ward occupancy

Inpatient Bed Census & Ward Occupancy Hospital #01 (Mumbai Central Hub) Availability: 99.857% (MTBF 1,400h) Case Study Docs Dr. Priya Sharma

Inpatient Bed Census & Ward Occupancy
Real-time bed tracking across hospital wards to prevent double allocations.

TOTAL WARD BEDS 140 Across 4 hospital wings
OCCUPIED BEDS 112 80.0% Current Occupancy
AVAILABLE BEDS 22 Ready for immediate admission
SANITIZATION / MAINTENANCE 6 Housekeeping turnover

Intensive Care Unit (ICU) Live Bed Grid

Bed #	Unit	Patient Name	UHID	Admitted Date	Attending Specialist	Bed Status
ICU-01	Intensive Care	Kishore Varma	AH-2026-0112	2026-10-01	Dr. Priya Sharma	Occupied (Critical)
ICU-02	Intensive Care	Fatima Sheikh	AH-2026-0814	2026-10-01	Dr. Suresh Deshmukh	Occupied (Ventilated)
ICU-03	Intensive Care	-	-	-	-	Under Housekeeping
ICU-04	Intensive Care	Rajesh Kumar	AH-2026-0982	2026-10-02	Dr. Priya Sharma	Occupied (Monitoring)
ICU-05	Intensive Care	-	-	-	-	Available for Triage

Inpatient bed census and ward occupancy

What it shows: Totals of 140 beds across four wings (112 occupied, 22 available, 6 in housekeeping) and a live ICU bed grid listing patient, admitting date and responsible specialist.

Why it matters: Supports the IPD workflow and prevents double allocation of beds. The same patient (ICU-04) appears across screens, showing the single shared record.

Laboratory information system

Laboratory Information System (LIS) Hospital #01 (Mumbai Central Hub) Availability: 99.857% (MTBF 1,400h) Case Study Docs Dr. Priya Sharma

Laboratory Information System (LIS)
Diagnostic test accessioning, reference range validation, and panic value escalation.

Laboratory Orders & Panic Value Notifications

Order ID	Patient UHID	Test Name	Result	Reference Range	Clinical Flag	Status
LAB-9012	AH-2026-0982 (Rajesh Kumar)	High-Sensitivity Troponin I	184.2 ng/L	< 14.0 ng/L	PANIC VALUE: Acute Myocardial Injury	Verified & Reported
LAB-9013	AH-2026-0982 (Rajesh Kumar)	Serum Potassium (K+)	4.1 mEq/L	3.5 - 5.1 mEq/L	Normal Range	Verified & Reported
LAB-8994	AH-2026-1184 (Amit Patel)	Complete Blood Count (WBC)	14.2 x10 ³ /uL	4.0 - 11.0 x10 ³ /uL	Mild Leukocytosis	Verified & Reported

Laboratory information system

What it shows: Lab orders with result, reference range and clinical flag. A Troponin I of 184.2 ng/L against a normal below 14.0 is flagged as a red panic value (acute myocardial injury).

Why it matters: Covers the lab-orders requirement and shows automatic escalation of critical results, which is another patient-safety function beyond allergies.

Pharmacy barcode verification and dispensing

Pharmacy Barcode Dispensing Station Hospital #01 (Mumbai Central Hub) Availability: 99.857% (MTBF 1,400h) Case Study Docs Dr. Priya Sharma

Pharmacy Barcode Verification & Dispensing
Secondary physical safety gate ensuring dispensed medication matches authorized electronic prescription.

Awaiting Medication Barcode Scan
Scan medication package barcode before issuing to ward nursing staff. System will cross-check batch expiry and drug contraindications.

Scan Barcode: Azithromycin 500mg (Batch AZ-9914) Complete Dispense & Decrement Stock

Dispensing Queue (Hospital #01 Dispensary)

Rx ID	Patient UHID	Prescribed Medication	Safety Status	Verification
RX-2026-7841	AH-2026-0982 (Rajesh Kumar)	Azithromycin 500mg Oral	Safety Cleared (Macrolide)	Awaiting Scan
RX-2026-7102	AH-2026-0982 (Rajesh Kumar)	Atorvastatin 40mg Oral	Dispensed & Verified	Verified by Rohan Mehta

Pharmacy barcode verification and dispensing

What it shows: The dispensing queue shows each prescription's safety status. The pharmacist scans the medicine barcode (Azithromycin, batch AZ-9914) before issuing; the system checks batch expiry and contraindications and then reduces stock.

Why it matters: A second, physical safety gate. Even if a wrong drug reached the pharmacy, the barcode check must match the authorised e-prescription.

Discharge summary

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MediTrack

AAROGYA HEALTHCARE

CONNECTED HEALTHCARE FACILITY

Hospital #01 (Mumbai Central Hub)

CLINICAL CORE

Main Dashboard

Patient Directory & History

Patient Registration

OPD Clinical Notes (SOAP)

CPOE Prescribing

Inpatient Bed Census

Laboratory (LIS)

Pharmacy Dispensing

Discharge Summary

GOVERNANCE & OPERATIONS

Chained Audit Ledger

2.4M Legacy Migration

Risk & Safety Center

PS

Dr. Priya Sharma

Attending Physician

Inpatient Discharge Coordination Protocol

Hospital #01 (Mumbai Central Hub)

Availability: 99.857% (MTBF 1,400h)

Case Study Docs

Dr. Priya Sharma

Inpatient Discharge Summary Protocol

Standardized continuity of care documentation preventing post-discharge adverse events.

Patient Name

Admission Date

Anticipated Discharge

Rajesh Kumar (AH-2026-0982)

2026-10-02 08:30

06/10/2026

Final Discharge Diagnoses

Acute Coronary Syndrome, Unstable Angina, Essential Hypertension

Hospital Course Summary

Patient admitted with substernal chest pain. Cardiac markers monitored. Managed medically. Hemodynamically stable. Documented severe Penicillin allergy noted on permanent summary card.

Reconciled Discharge Medications (Verified Free of Allergens)

1. Azithromycin 500mg OD x 3 days. 2. Atorvastatin 40mg OD at night. 3. Aspirin 75mg OD post-meals. CONTRAINDICATED: Do NOT administer Penicillins or Beta-Lactams.

Save Draft

Sign & Generate Patient Summary PDF

Discharge summary

What it shows: A form pre-filled for the patient: diagnoses, hospital course (noting the Penicillin allergy), and reconciled discharge medications marked as verified free of allergens, with an explicit contraindication warning for penicillins and beta-lactams.

Why it matters: Carries the allergy information forward to whoever treats the patient next, so the risk does not reappear after discharge.

Cryptographically chained audit ledger

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MediTrack

AAROGYA HEALTHCARE

CONNECTED HEALTHCARE FACILITY

Hospital #01 (Mumbai Central Hub)

CLINICAL CORE

Main Dashboard

Patient Directory & History

Patient Registration

OPD Clinical Notes (SOAP)

CPOE Prescribing

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Laboratory (LIS)

Pharmacy Dispensing

Discharge Summary

GOVERNANCE & OPERATIONS

Chained Audit Ledger

2.4M Legacy Migration

Risk & Safety Center

PS

Dr. Priya Sharma

Attending Physician

Cryptographically Chained Audit Ledger (SHA-256)

Hospital #01 (Mumbai Central Hub)

Availability: 99.857% (MTBF 1,400h)

Case Study Docs

Dr. Priya Sharma

Cryptographically Chained Audit Ledger (SHA-256)

Append-only Merkle chain providing legal non-repudiation and tamper detection.

Verify Cryptographic Chain Integrity

Legal Evidentiary Rule: Zero Silent Record Modification

Every patient chart view, prescription order, lab requisition, and override is cryptographically hashed with its preceding log block: $\text{Hash}(n) = \text{SHA-256}(\text{Hash}(n-1) + \text{Timestamp} + \text{User} + \text{Payload})$.

Recent Chained Audit Blocks

Block #	UTC Timestamp	User / Role	Action Category	Target UHID	Current SHA-256 Hash	Previous Hash
#42912	2026-10-02 10:14:22	Dr. Priya Sharma (MD)	SAFETY_INTERLOCK_HARD_STOP	AH-2026-0982	9f83c18b...e4a2	7d28aa19...b184
#42911	2026-10-02 09:30:14	Dr. Priya Sharma (MD)	RX_ISSUED_EVENT	AH-2026-0982	7d28aa19...b184	33c914e0...88ab
#42910	2026-10-02 08:35:01	Nurse Anjali Patil	EMERGENCY_TRIAGE_VITALS	AH-2026-0982	33c914e0...88ab	12af0092...5512
#42909	2026-10-02 08:31:18	Nurse Anjali Patil	CHART_ACCESSED	AH-2026-0982	12af0092...5512	009c8812...3319

Cryptographically chained audit ledger

What it shows: A table of audit blocks, each with timestamp, user and role, action category, patient ID, its own SHA-256 hash and the previous block's hash. The block formula is shown: $\text{Hash}(n) = \text{SHA-256}(\text{Hash}(n-1) + \text{Timestamp} + \text{User} + \text{Payload})$. The hard-stop event at 10:14:22 matches the Amoxicillin interception on the dashboard.

Why it matters: This is the tamper-evident ledger (extra feature 5.3). Because each hash depends on the one before it, any edit to history breaks the chain, and the "Verify Cryptographic Chain Integrity" button re-checks it.

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Legacy data migration and quarantine workspace

The screenshot shows the 'Legacy Data Migration & Quarantine Workspace' interface. At the top, it displays 'Hospital #01 (Mumbai Central Hub)' and 'Availability: 99.857% (MTBF 1,400h)'. Below this, a summary section shows 'TOTAL LEGACY RECORDS' as 2,400,000 (2 FoxPro, 3 Excel, 1 Paper site), 'CLEAN INGESTED (93%)' as 2,232,000 (Validated & loaded into live tables), 'QUARANTINE ISOLATED (7%)' as 168,000 (Zero unverified rows in live tables), and 'RECONCILED TO DATE' as 12,410 (Verified by Medical Custodians). A 'Quarantine Review Queue (168,000 Problematic Records)' table lists specific records with their source systems, legacy IDs, patient names, error classifications (e.g., ERR_DATE_MALFORMED, ERR_ID_COLLISION), raw payload issues, and actions (e.g., Reconcile Date, Resolve Collision, Convert Serial).

Legacy data migration and quarantine workspace

What it shows: Of 2,400,000 legacy rows, 2,232,000 (93%) were validated and loaded and 168,000 (7%) were isolated in quarantine. The review queue lists each problem record with its source system, an error class (for example malformed date, ID collision) and a repair action such as Reconcile Date or Resolve Collision.

Why it matters: This is extra feature 5.4. Bad rows never reach the live tables, yet none are lost: each can be corrected and re-introduced by a medical custodian.

Clinical risk and safety governance centre

The screenshot shows the 'Clinical Risk & Safety Governance Center' interface. It displays 'Hospital #01 (Mumbai Central Hub)' and 'Availability: 99.857% (MTBF 1,400h)'. Below this, a 'Clinical Risk Management & RMMM Matrix' section shows a 'Formal 5x5 Probability-Impact matrix with dedicated RMMM plans for top clinical hazards.' A table titled 'Top 3 Project Risks & RMMM Mitigation Plans' lists three risks: R-01 (Fatal Prescribing Allergy Incident), R-02 (Legacy Migration Data Corruption), and R-03 (Audit Tampering / Legal Failure), each with its probability, impact, exposure, and a detailed mitigation and management plan (RMMM).

Clinical risk and safety governance centre

What it shows: A 5x5 probability-impact matrix with the top three risks and their RMMM plans: fatal allergy incident (exposure 20), legacy migration corruption (exposure 20) and audit tampering (exposure 15).

Why it matters: Shows that risk management is built into the system's governance view, and that the top three risks each map to one of the extra features.

User account profile and credentials

MediTrack
AAROGYA HEALTHCARE

CONNECTED HEALTHCARE FACILITY
Hospital #01 (Mumbai Central Hub)

Patient Registration
OPD Clinical Notes (SOAP)
CPOE Prescribing **ALLERGY GATE**
Inpatient Bed Census
Laboratory (LIS)
Pharmacy Dispensing
Discharge Summary
GOVERNANCE & OPERATIONS
Chained Audit Ledger
2.4M Legacy Migration **7% QUARANTINE**
Risk & Safety Center
User Account Profile
ENGINEERING DELIVERABLES
Deliverables & Documents Hub

PS Dr. Priya Sharma
Attending Physician

User Account Profile & Credentials Hospital #01 (Mumbai Central Hub) Availability: 99.857% (MTBF 1,400h) Case Study Docs Dr. Priya Sharma

User Account Profile & Clinical Credentials
Authenticated clinician identity, regulatory license verification, and session governance. [Sign Out of MediTrack](#)

Clinical Identity & Licensure

Full Legal Name	Dr. Priya Sharma
Assigned Role	Attending Physician
State Medical Council License	MH-MED-2018-0914 (Verified Active)
Clinical Department	Cardiology & Emergency Medicine
Primary Healthcare Facility	Hospital #01 (Mumbai Central Hub)
Staff Credential Badge ID	STAFF-9021-CARDIOLOGY
Current Shift Status	Active Duty (08:00 - 16:00 IST)

Security Credentials & Session Tokens

Authentication Protocol	OAuth 2.0 / OpenID Connect with PKCE
Multi-Factor Authentication (MFA)	Enforced: FIDO2 Hardware Key
Current Session Token	JWT-SEC-9982410-EXP15M
Session Idle Timeout	5 Minutes (Automatic Terminal Lock)
Prescribing Authorization	Level 4: Full Formulary CPOE
Audit Attribution	Cryptographically chained to SHA-256 blocks

Switch Simulation User Role:
 Attending Doctor Staff Nurse Chief Pharmacist Chief Medical Officer

User account profile and credentials

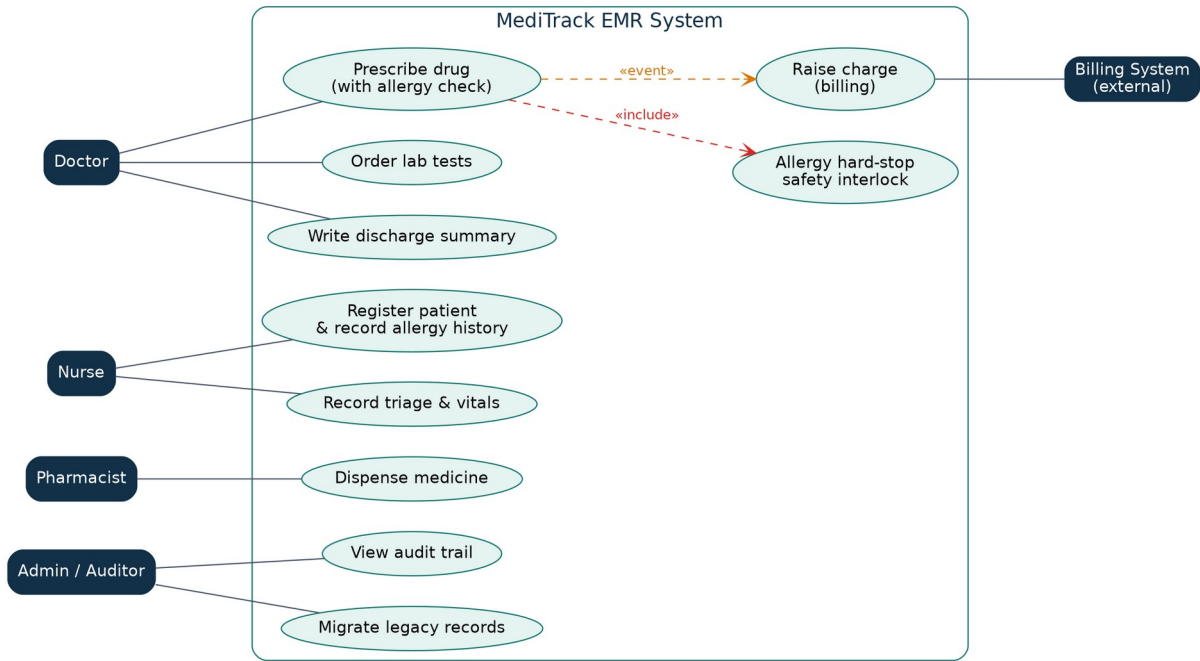
What it shows: Clinical licence, role, department and shift status alongside security settings: OAuth 2.0 / OpenID Connect with PKCE, hardware-key multi-factor authentication, a short-lived session token, a 5-minute idle lock, and a prescribing authorisation level.

Why it matters: Demonstrates the security and access-control non-functional requirements, and that only licensed, verified clinicians can prescribe.

7. Architecture and UML

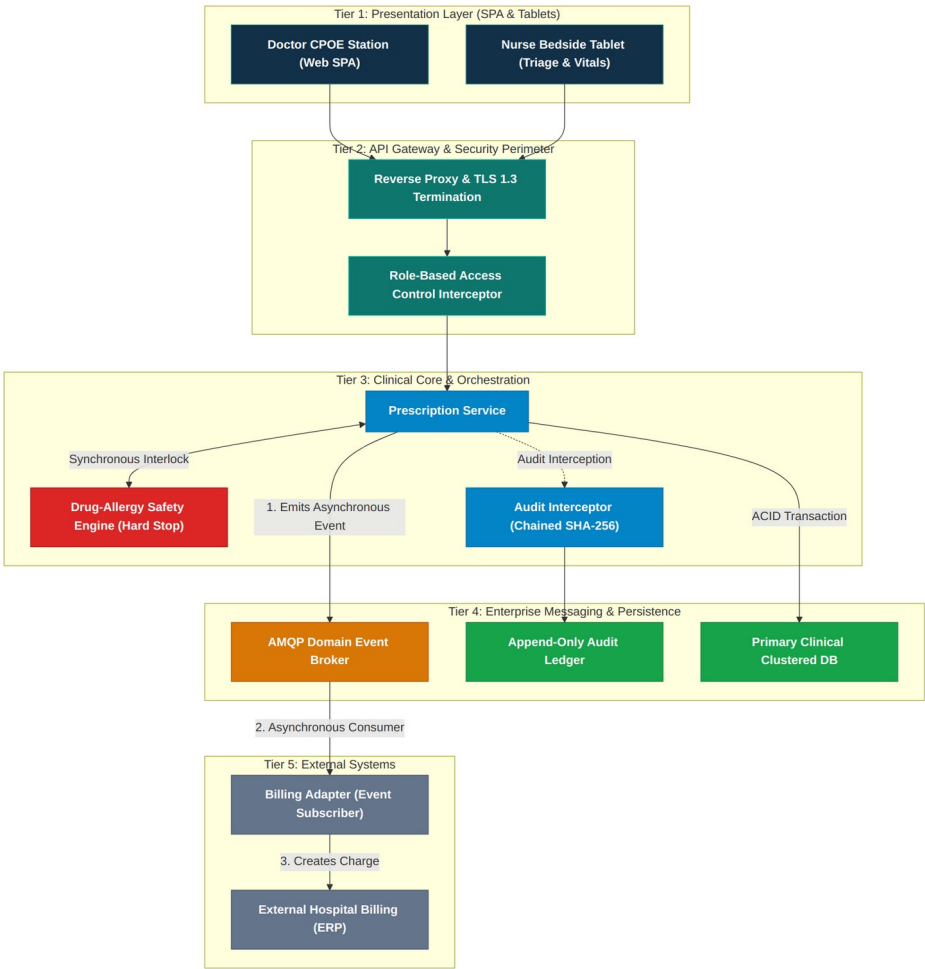
7.1 Use case diagram

Five actors interact with the system. The prescription use case always includes the safety interlock, and raises a billing event that the external billing system consumes asynchronously.



Use case diagram: actors and principal use cases

7.2 Five-tier software architecture

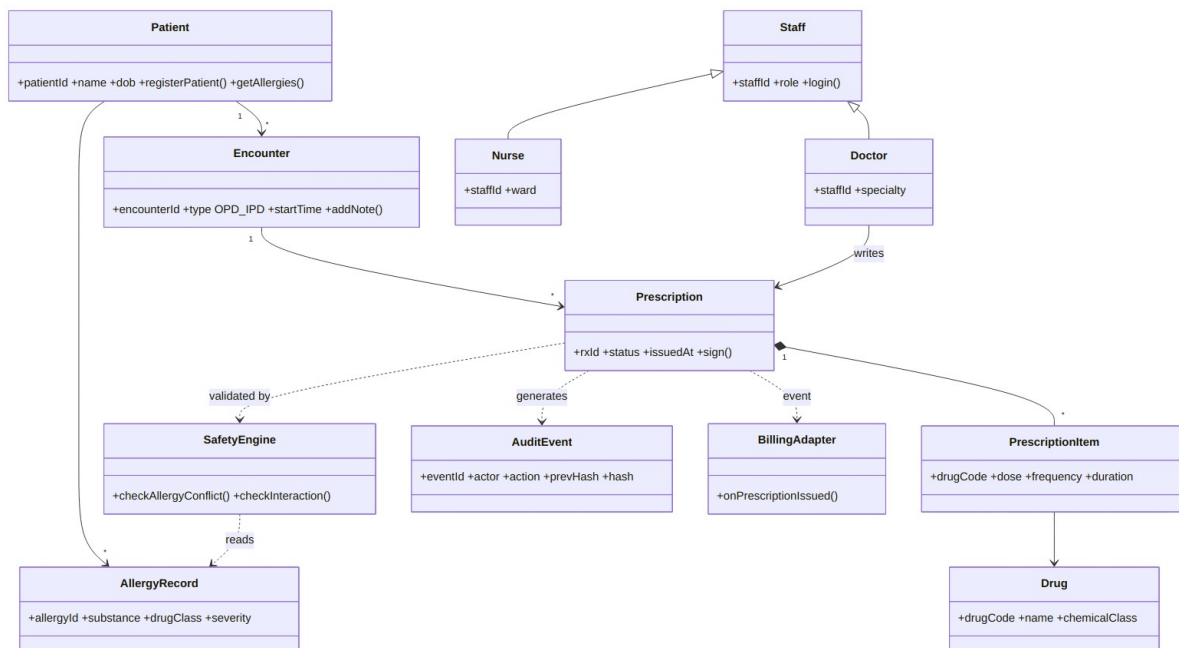


Five-tier architecture with synchronous safety check and asynchronous billing

Tier	Responsibility
1. Presentation	Doctor CPOE web SPA and nurse bedside tablet.

Tier	Responsibility
2. Gateway and security	TLS 1.3 termination, OAuth 2.0 / OIDC with JWT sessions, multi-factor authentication, role-based access control.
3. Clinical core	Prescription Service, Drug-Allergy Safety Engine (hard stop), Audit Interceptor.
4. Messaging and persistence	AMQP event broker, clinical database, append-only audit ledger.
5. External systems	Billing Adapter (event subscriber) and the external hospital billing ERP.

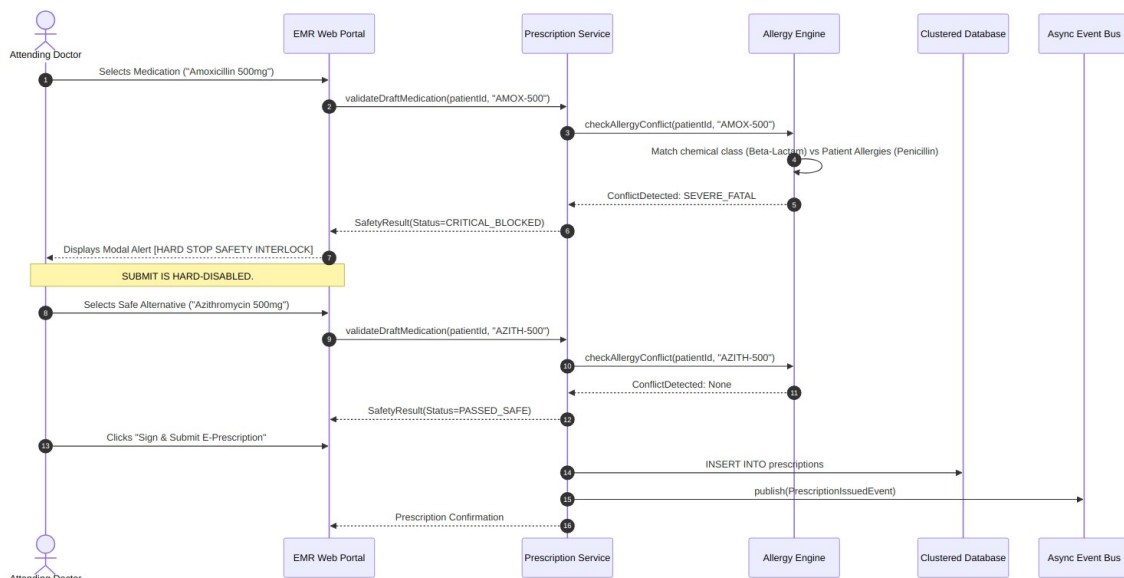
7.3 Class diagram



Class diagram: core domain model of the prescribing module

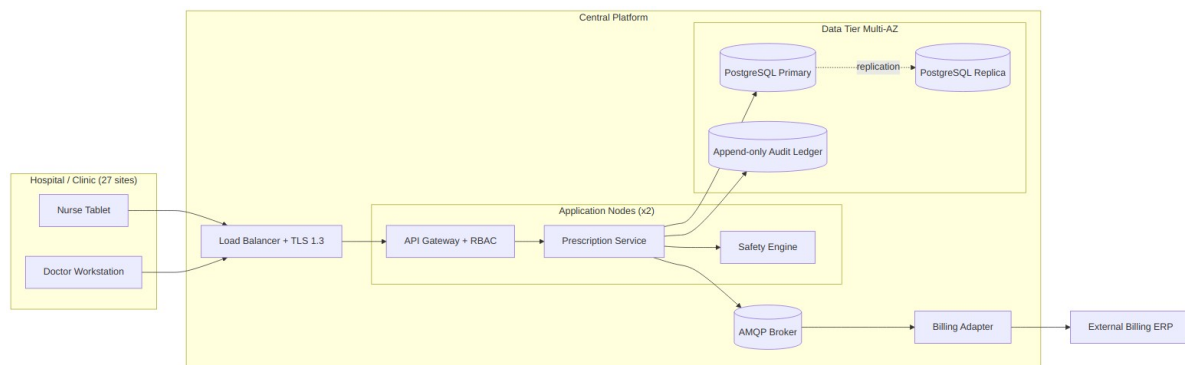
7.4 Sequence diagram: prescribe drug with allergy check

This is the scenario that prevents the 2024 incidents. The doctor first picks Amoxicillin; the engine matches its beta-lactam class against the recorded Penicillin allergy and blocks submission. The doctor then chooses Azithromycin, which passes, and only then can the prescription be signed, stored, and announced to billing.



Sequence diagram: allergy check hard stop

7.5 Deployment diagram



Deployment view: sites, application nodes, multi-AZ data tier and billing integration

8. How the backend works

When a doctor writes a prescription, the following happens, in this order:

1. **Request.** The request reaches the API Gateway over TLS 1.3 with a signed session token (OAuth 2.0 / OpenID Connect, JWT); the RBAC interceptor confirms the caller is allowed to prescribe.
2. **Safety check (synchronous).** The Prescription Service asks the Drug-Allergy Safety Engine to validate each drug against the patient's allergy record. A conflict returns CRITICAL_BLOCKED and the flow stops here.
3. **Commit.** If safe, the prescription is committed in a single ACID transaction to the Multi-AZ PostgreSQL database.
4. **Audit.** The Audit Interceptor appends a hash-chained entry to the append-only ledger.
5. **Event (asynchronous).** A PrescriptionIssuedEvent is published to the AMQP bus.
6. **Billing.** The Billing Adapter consumes the event and creates the charge in the external billing system, without ever blocking the doctor.

The key design idea

The step that must be **immediate and certain** (the safety check) is synchronous. The step that can be **eventual and tolerant of failure** (billing) is asynchronous. Each dependency is placed on the side of the

line that matches how much clinical risk it carries.

9. Quality assurance: how testing would have caught the 310 incidents

The 310 incidents had one cause: allergy history was missing or not consulted at the moment of prescribing. The test design attacks that cause at three levels.

Technique	What it covers for this project
Decision table testing	Every combination of allergy present or absent, same-class or cross-class drug, and severity is a row; each row has an expected action (PASS, BLOCK or HARD STOP).
Boundary Value Analysis	Dosage fields are tested at minimum, just below minimum, just above minimum, nominal, just below maximum, maximum and just above maximum, to catch out-of-range doses.
Migration regression suite	After each load: row counts reconcile (loaded + quarantined = extracted), no duplicate IDs, all dates valid, checksums match.
Traceability matrix	Every functional requirement links to at least one test case, so no safety requirement can be left untested.

Metric	Target
Defect Removal Efficiency (DRE)	At least 92% before go-live. $DRE = \text{defects found before release} / (\text{defects found before release} + \text{defects found after release})$
MTBF	1,400 hours
MTTR	2 hours
Implied availability	$MTBF / (MTBF + MTTR) = 1,400 / 1,402$, about 99.86%

The prototype dashboard reports a measured **DRE of 93.88%**, above the 92% target, and an availability of **99.857%** (MTBF 1,400 h, MTTR 2 h).

10. Project management summary

The full artefacts are in the companion documents. The summary below states the headline decisions and where to find the evidence.

Area	Summary
Effort estimation	COCOMO estimate for 46 KLOC with the mode justified for a medium-experienced team on regulated clinical software; sanity-checked against Brooks's Law. The model does not capture migration or training effort, which are estimated separately.
Planning	Four-level WBS, network diagram with critical path, Gantt chart and resource histogram, fitted inside the 18-month ceiling.
Risk	Risk register on a 5x5 probability-impact matrix. Top three: R-01 fatal prescribing allergy incident ($4 \times 5 = 20$, critical), R-02 legacy migration data corruption ($4 \times 5 = 20$, critical), R-03 audit tampering / legal failure ($3 \times 5 = 15$, high). Each has an RMMM plan: the 14-rule hard stop; the 12-stage ETL with quarantine and a scripted 24-minute rollback; and the append-only SHA-256 ledger with write-once archival.

Area	Summary
DevOps	CI/CD pipeline with a staged clinical-safety gate. Steps that must never be automated in a regulated environment: approval to release into production, changes to the clinical rule tables without clinical sign-off, promotion of migrated data, and any edit to the audit ledger.

11. Objectives versus outcomes

Objective in the case study	How this project meets it	Where to see it
Justify process model	V-Model chosen; Waterfall and Scrum rejected with reasons	Section 3
12 FRs and 8 measurable NFRs, MoSCoW	Elicited and prioritised; every NFR is measurable	SRS
Use case, class, sequence, deployment diagrams	Produced and kept consistent with one another	Section 7; UML package
Prescription module loosely coupled from billing	Port-and-Adapter plus AMQP events	Section 5.2, 7.2
BVA, decision table, migration regression suite	Designed and linked to requirements	Section 9; Test Plan
WBS, Gantt, risk register, COCOMO	Delivered against the 18-month ceiling	Section 10; Project Plan
CI/CD with clinical-safety gate	Defined, with explicit non-automatable steps	Section 10

12. Limitations and future scope

Being clear about limits is part of safety engineering. The following are known and deliberate boundaries of this submission.

- **The UI is a prototype.** It demonstrates workflow and design intent; the backend described here is a design, not a deployed production system.
- **The audit ledger is tamper-evident, not tamper-proof.** Someone with full control could rebuild the whole chain. Future work: periodically anchor the latest hash with an external party.
- **Event publishing reliability.** Committing to the database and publishing to the broker are two separate actions. Future work: a transactional outbox so no event can be lost between them.
- **Drug knowledge base.** The cross-class rules need clinical review and a maintained drug database before real use.
- **Regulatory scope.** For deployment in India, the Digital Personal Data Protection Act 2023 and national digital health guidelines would be assessed alongside HIPAA-style controls.
- **Further ideas:** drug-drug interaction checks, offline mode for rural clinics, patient-facing portal, analytics on near-miss events.